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Design Rationale

Learning goals were developed after drafting and refining the core messages for the project. The two messages that we settled on were (1) water is a precious resource, and (2) humans have solved and can solve problems related to local water issues, using technology. The learning goals were developed with the core audience of Grade 6-12 students in mind. The four learning objectives that we developed were:

- Learners **identify** historical uses of water by humans.
- Learners **examine** artifacts to **relate** historical water use to technological solutions.
- Learners use interactive models to **reflect** on their own water use.
- Learners **connect** solutions to water use problems to issues in their local community.

Given constraints of time and technology, we were able to focus more on some of the objectives than others, but the intent was to connect visitors to historical and contemporary water use and learn how innovative technology can help solve the ‘wicked problem’ of water resource management and scarcity.

We designed a circular spatial layout to encourage open exploration of ideas and all call to mind the concept of the water cycle and how technologies are adapted through time, from ancient to modern contexts. We grouped artifacts and presentations into three main areas focusing on irrigation, transportation, and filtration and designed a central meeting/classroom area to accommodate groups such as school class visits, or virtual presentations. We planned an interactive reflection wall (not fully prototyped) as a way for visitors to reflect on their experience in the exhibit and connect the issue to their lives and their communities.

We placed 3D artifacts and image-and-text pairings along the walls of the circular space with an infographic on the floor to ground visitors to the issue of global water use. The planned “Where do you get your water?” wall, while not yet interactive, anchors visitors to our fourth learning objective to connect them to local water issues in their community. This reflection zone is our main transfer mechanism, as it gently pushes visitors to shift from learning about distant case studies to considering the sources and infrastructures that make their own water use possible.

Learning Science Integration

We used backward design by thinking about the big ideas we wanted visitors to take away from the exhibit and designing learning objectives around those ideas. Although there are not assessments in the experience per se, we thought about how to encourage visitors to self-reflect on their relationship to water within their communities. We then thought about the activities and content that could help visitors meet those objects, but using multimodal artifacts, reflective questions, and an intentionally designed space to encourage discussion and questioning.

As our primary audience is 6-12th graders, we adapted text passages to be age- and grade-level appropriate, adjusting vocabulary, volume of text, and complexity of language to limit cognitive overload. The 3D objects were designed to promote deep processing by making abstract ideas like changes in filtration technology *physically* present in the space. Finally, the water reflection wall and benches are meant to ground all the water concepts learned in the user's personal experience. Prompts such as "Where do you get your water?" ask visitors to connect exhibit content to their own homes and communities, which is critical for transfer beyond the museum. Our original plan was to engineer an interactive post-it wall that creates a sea of responses to this question, so exhibit viewers can see the scale of how water affects everyone around them.

Individual Contribution

As a current and former teacher and curriculum developer, I took the lead on turning our initial conversation about core messages and big ideas into four learning objectives. I tried to keep the number of objectives between 3-5, and use student and teacher-friendly language. I focused on achievable objectives but did not necessarily consider time or technology constraints, as the objectives were developed at the start of the project. I used action verbs that addressed various levels of complexity and Bloom's Taxonomy, and avoided general terms such as "understand" and "know."

I contributed to all of the group's weekly planning meetings and discussion and focused on the design of the space and editing and placement of assets within Spatial. I took an image from the MUSE group members that was a mock-up for the exhibit and turned it into a 3D object using meshy.ai. I then worked through several ideas and prototypes for the layout of the exhibit, gallery walls, and seating, collaborating with the group throughout. I troubleshooted and continuously updated the Spatial space as we neared project deadlines and received feedback.

Interaction and Reflection

We iterated on several ideas in the early phases of the project, collaborating with the MUSE team members. When they sent a prototype idea for the museum layout, I built several versions of a 3D model for the exhibit walls and presented them to the group for feedback. The MUSE members selected the model they liked best and that was aligned with their vision and the objectives of the exhibit. As our class members handled some of the technical aspects of the project, the MUSE members presented ideas that we tested and tried to determine if they were feasible, while contributing our own content ideas. During in-class group work we received some helpful advice and feedback from Luis and Andy on some of the technical aspects of the project, such as Unity integration and meshy troubleshooting. After our presentation, we received a lot of helpful feedback about the visitor experience, layout of the exhibit such as wayfinding and keeping visitors focused, and how to more authentically connect the visitors' experience to self-reflection. We were able to integrate a few small changes so far, such as adding a welcome screen and wayfinding, helping visitors to the exhibit when arriving in Spatial. A roof was also added to the exhibit to focus visitors' attention and experience more.

Collaboration Reflection

Overall the collaboration worked well with few roadblocks. We established norms over email and decided on Google Spaces to collaborate and do weekly calls for planning and progress updates. One challenge we found was the difference in project deadlines for the two classes. Although both deadlines felt relatively aggressive, the MUSE class' deadline was prior to ours, so it led to some stress and rushed changes or decisions that may have sacrificed quality in some aspects of the project. We also had some technical problems within Google that did not allow for seamless communication from the start. I would choose a different comms platform, such as Slack, next time for better collaboration. I would also have liked to add more to our physical space in terms of variety of interactions for the user. Because of the way we delegated the work, I think it made it more challenging to bring many new ideas to the table. We instead had ideas early on and stuck within those ideas and our "roles," without much flexibility. We also had several group members (including myself) with some personal/ work matters that did not allow for as much execution time as we would have liked.